

Regularized and Nonlinear Regression

Estimate continuous outcomes with interpretable baselines and robust evaluation.



Learning outcomes

01**Explain the bias-variance tradeoff.****02****Use regularization to stabilize coefficients.****03****Compare tree regressors with linear models.**

The concept system

01

regularization

02

bias and variance

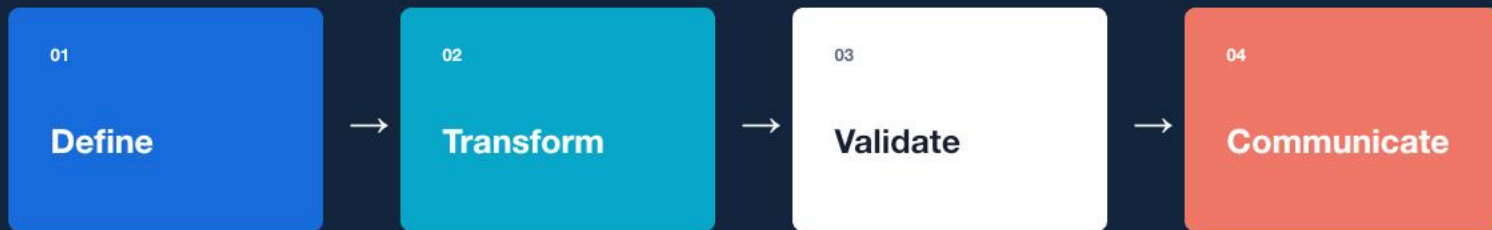
03

trees

04

cross-validation

A repeatable reasoning loop



Compare ridge, lasso, and gradient boosting on house prices.

Guided lab

LAB BRIEF

Compare ridge, lasso, and gradient boosting on house prices.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A model comparison table and justified selection.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Regression Model Card

Evaluation 30%, diagnostics 25%, model choice 20%, reproducibility 15%, communication 10%.

What should remain after the lesson?

CONCEPT

regularization + bias and variance

PRACTICE

Compare ridge, lasso, and gradient boosting on house prices.

EVIDENCE

A model comparison table and justified selection.
